

Question Paper

Exam Date & Time: 26-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026

SUBJECT: CHM 1003/CHM_1003 APPLIED CHEMISTRY FOR ENGINEERS

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) i) Explain the construction and working of lead acid battery with appropriate discharging chemical reactions and diagram. (4)
ii) A battery stores 360,000 joules of energy and discharges completely in 30 minutes. The mass of the battery is 5 kg. Calculate the power density (W/kg) of the battery.
(3+1 = 4 marks)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) The EMF of the Cadmium - Nickel cell is found to be 0.20 V at 298 K. (4)
i) What is the molar concentration of Cd ions in the solution, if Ni ion concentration is 2 M ?
ii) Write the reactions of anode and cathode.
iii) Calculate the change in free energy during the cell reaction at 298K.
(2+1+1 = 4 marks)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Differentiate between the fuel cell and Galvanic cell (Any 4 points). (2)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2A) Give reasons for the following: (4)
i) Polyvinylchloride is more crystalline than polystyrene.
ii) Chitosan is a polymeric biomaterial used for wound healing
iii) Ball milling method is not suitable for preparing polymer nanoparticles
iv) Organic light emitting diode (OLED) displays are superior to liquid crystal displays (LCD)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2B) Illustrate the mechanism of photo-responsive and electro-responsive polymers. (4)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) A sample of polypropylene has the degree of polymerization of 410, 550 and 600 was found to have the composition of 25%, 35% and 40% respectively. Calculate the number average and weight average molecular mass of it. (2)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3A) Explain the stepwise working of a continuous glucose monitoring (CGM) biosensor. (4)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 3B) A 50 mL sample of a monoprotic weak acid is titrated with 0.1000 M NaOH. The equivalence point is reached after adding 10 mL of NaOH. At the half-equivalence point the measured pH = 4.700. (a) Find K_a of the acid, (b) Find the initial concentration of the acid, (c) Calculate the pH after adding 5 mL NaOH. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3C) Discuss the construction of receptor used in catalytic bead sensor. How does poisoning of the catalyst affect the operation of hydrogen sensors? (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4A) i) List the potential health effects of beryllium and cadmium exposure. (4)
ii) Explain how recycling and extended producer responsibility can reduce electronic waste.
iii) Mention any two common practices of pre-treatment of e-waste.
(1+2+1 = 4 marks)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4B) i) Describe soil pollution through improper e-waste disposal. (4)
ii) Differentiate between recycling and circular economy.
(2+2 = 4 marks)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 4C) Discuss the specific safety measures and practices that manufacturers incorporate to reduce the hazards associated with lithium-ion batteries. (2)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 5A) i) Name the electrode used in the determination of pK_a of weak acid. (4)
ii) What is the working principle of flame photometry?
iii) A solution of a colored compound has a concentration of 0.006 mol/L. When a light of wavelength 540 nm passes through a 1 cm cuvette containing the solution, the absorbance is measured as 0.95.
(a) Calculate the molar absorptivity (ϵ) of the compound at 540 nm.
(b) What will be the absorbance if the concentration is increased to 0.02 mol/L using the same cuvette and wavelength?
(1+1+2 = 4 marks)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 5B) Differentiate between - (4)
i) PVD and CVD techniques.
ii) Micro and ultra-filtration membranes.
(2+2 = 4 marks)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 5C) Explain the mechanism of Electro-responsive polymers and their applications. (2)

Course Outcome: "CO2" / Cognitive Level: "Understand"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	12
2	2	CO 2	16
3	3	CO 3	14
4	4	CO 4	8
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-
2026

CHM_1003:APPLIED CHEMISTRY FOR
ENGINEERS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	8	24(48%)
3	Application	6	24(48%)
4	Analysis	1	2(4%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)